Andrew Zheng

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Experience

Software Engineer Intern | Miovision | Waterloo, ON — May 2026 – Aug 2026

- Re-engineered RAUC firmware update path for Yocto-based x86-64 Linux devices in Node.js and AWS IoT via adaptive block hash index HTTP streaming to reduce OTA bandwidth cost by **98%** (1.4GB to 21MB)
- Built end-to-end HIL release pipeline for embedded ARM64 Linux devices in Node.js and AWS/GitLab on Terraform-provisioned, Ansible-configured OTA runners to poll **15** health metrics over **48h** with **<1%** frame-drop
- Deployed 360° camera emulator app to embedded ARM64 Linux devices on Firecracker microVM in Go/Docker to virtualize H.264 RTSP streaming + camera API, enabling camera-free HIL development
- Architected fleet-wide dual lookup for device search engine in Kotlin/Spring Boot to stream schema registry Kafka data to Kubernetes-hosted index, eliminating **100%** of stale results across **43K+** devices

Embedded Systems Developer | Waterloo Rocketry | Waterloo, ON — Sep 2025 – Present

- Designed 4-layer recovery parachute PCB for the **world record highest amateur liquid rocket** (**63,497 ft**), integrating dual 9V LiPo power stages with StratoLogger altimeters and Rocket CAN bus
- Engineered hardware fail-safe for PIC18 microcontroller to drive a P/N-channel MOSFET network to arm dual altimeters, fire pyros, and guarantee parachute deployment at peak altitude

Founder & Case Coach | Case Tutoring Startup | Toronto, ON — May 2024 – Aug 2025

- Drove **$10K+ revenue** through GTM campaign for self-founded business case contest coaching service, scaling across **150+** high school students from **10+** schools
- Won **1st Place** globally and coached **50+** qualifiers/**30+** international finalists at DECA ICDC, America's largest high school business case contest (230K+ members)

Projects

Allocate | macOS CPU/Thermal Fleet Controller | *Rust, XNU, Mach IPC, Prometheus, Grafana* — GitHub

- Built system-level macOS CPU governor using XNU QoS and POSIX policies to steer workloads across P/E cores and reduce background CPU consumption by **79%** with **100%** app uptime
- Engineered Rust FFI telemetry daemon to sample CPU, AppleSMC, IOReport, DVFS, and SoC power metrics every **500ms**, displayed on native AppKit live CPU chart
- Built 18-sensor Rust PID-based thermal controller reading CPU/GPU/ANE to dynamically adjust E-core policies through Mach/XPC, reducing CPU temperature by **12°C/22°F** without hardware fans
- Shipped native end-to-end SwiftUI app and Prometheus/Grafana fleet observability pipeline with configurable thresholds for per-app CPU controls, thermal, power, and throttling analysis

GitIRL Bot | Hack the North 2026 Winner | *Python, FastAPI, SLAM, OpenAI API* — Devpost

- Won Best use of Bracket Bot Hardware at Canada's Biggest Hackathon by building an autonomous version control system to map world states and command a humanoid robot to restore environment changes
- Designed fleet orchestration pipeline across Bracket Bot and cloud platform, indexing **6.9M+** telemetry points in Elasticsearch to enable hybrid semantic search processing for robotic actions

CrawlStars | Distributed Search Engine | *Golang, JavaScript, MongoDB, REST API* — GitHub

- Engineered concurrent crawlers in producer-consumer architecture to process **2000+ pages**/min in **<15MB**
- Optimized deduplication to enable O(1) lookup for **50K+** URLs and filter **70%** of redundant crawls
- Deployed search engine to calculate SEO relevance via MongoDB Atlas aggregation pipeline for **50K+** URLs

Skills

Languages: C, C++, Python, Rust, Go, Java, Kotlin, Swift, JavaScript, TypeScript, HTML, CSS, Ruby, Verilog
Infrastructure: AWS, Linux, Docker, Kubernetes, Terraform, Ansible, Kafka, CI/CD, Git, Prometheus, Grafana
Backend: Node.js, Express.js, FastAPI, Spring Boot, Ruby on Rails, REST API, MongoDB, Jupyter
Frontend/Other: React, Next.js, Tailwind, Streamlit, XGBoost, SLAM, VLA, Ubuntu, Mach, Apple IOKit

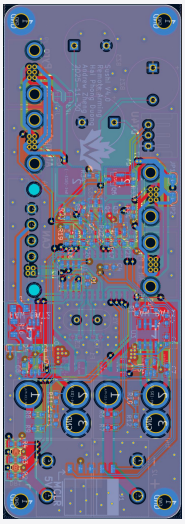
Education

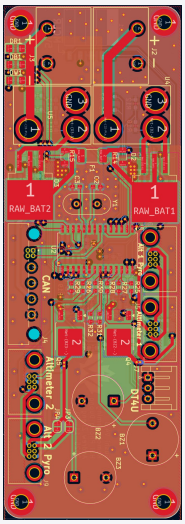
University of Waterloo | *BASc, Computer Engineering* — Sep 2025 – Apr 2030

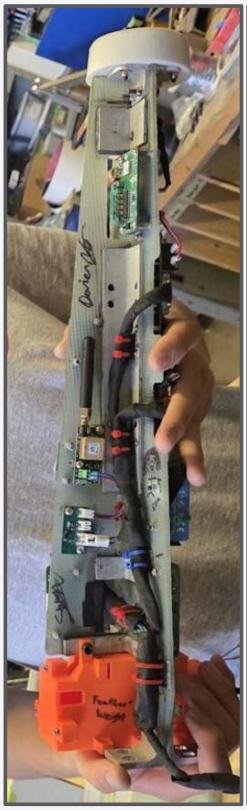
Recovery Arming PCB | Waterloo Rocketry 2026 [GitHub]

Portfolio - Andrew Zheng
[GitHub] [Website]

World Record Highest Amateur Liquid Rocket (63,497 ft)
Hardware: Safety-Critical KiCad PCB | Dual Power Input | P-MOS High-Side Mag Switch
Firmware: Embedded C (PIC18) | CAN Bus Telemetry Aggregation | ADC Monitoring

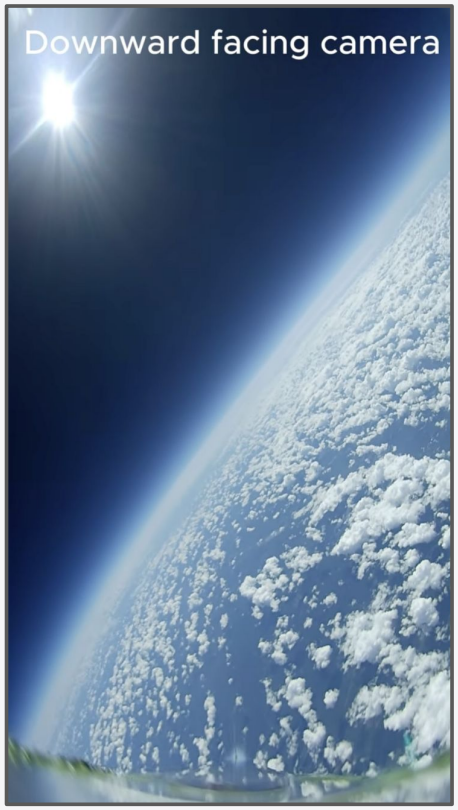




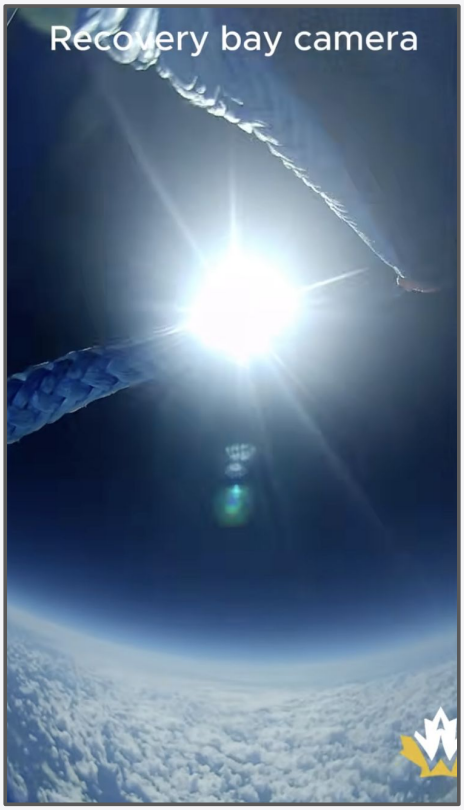




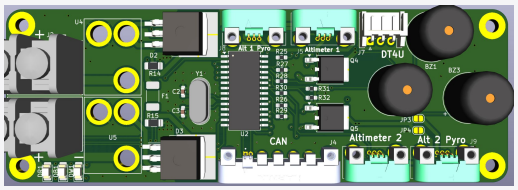
Electrical system integration

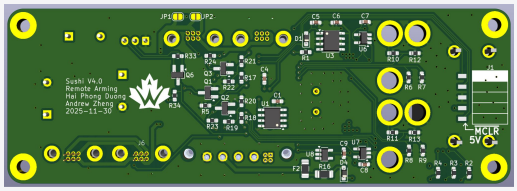


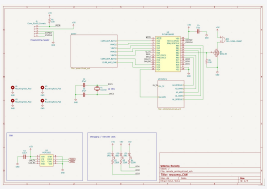
Rocket at peak altitude



Recovery parachute in action

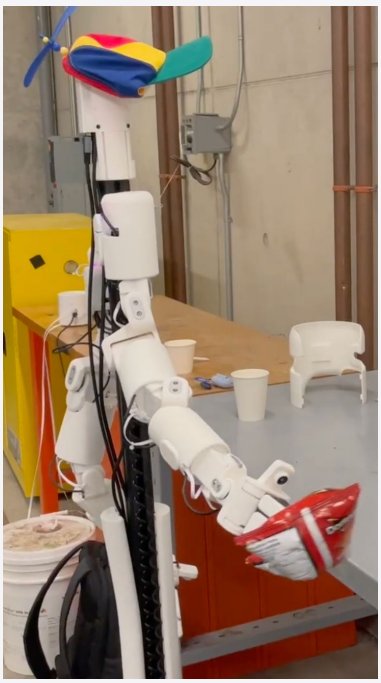






GitIRL Bot | Version Control for the Physical World [Devpost]

Hack the North 2026 Winner - Best use of Bracket Bot hardware
Python, FastAPI, React/Javascript, Docker, ESQL, OpenAI API | Humanoid VLA with 4D Memory

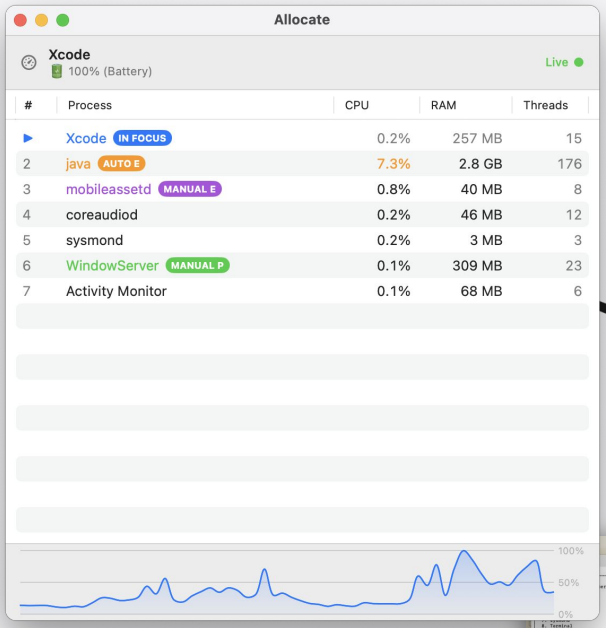




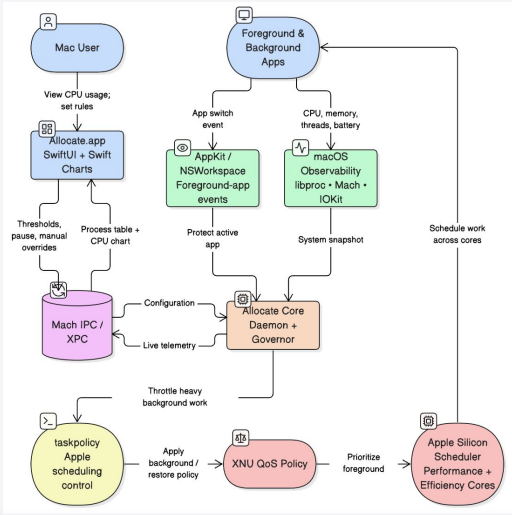
Allocate | Kernel-Level CPU Governor [GitHub] [Download]

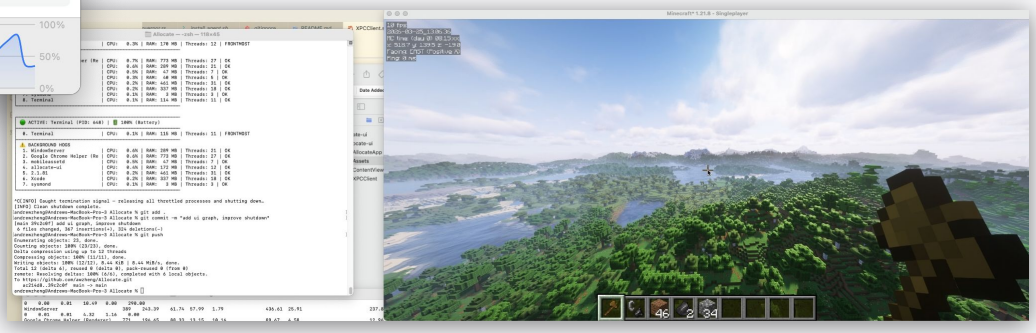
Portfolio - Andrew Zheng
[GitHub] [Website]

Rust/C FFI, XNU QoS, Mach IPC | 79% Reduced Background CPU Usage | 100% App Uptime





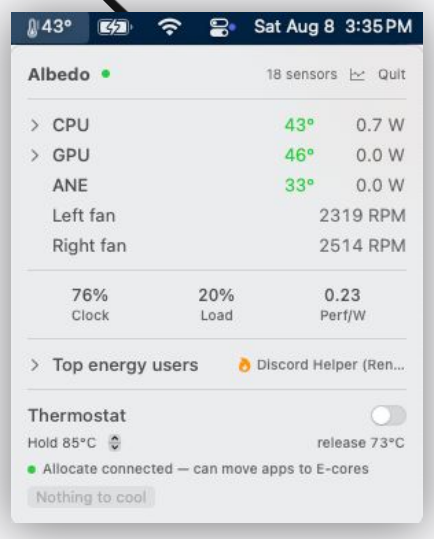


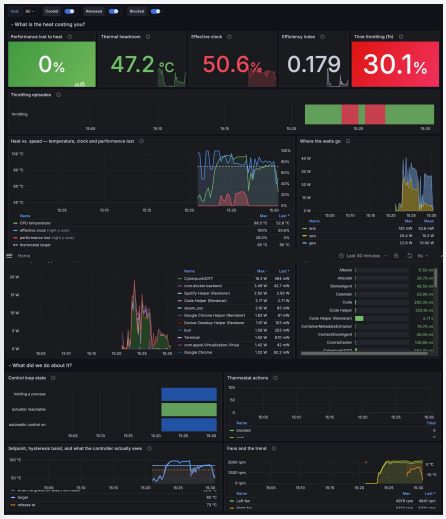


Designed XPC to control Allocate!

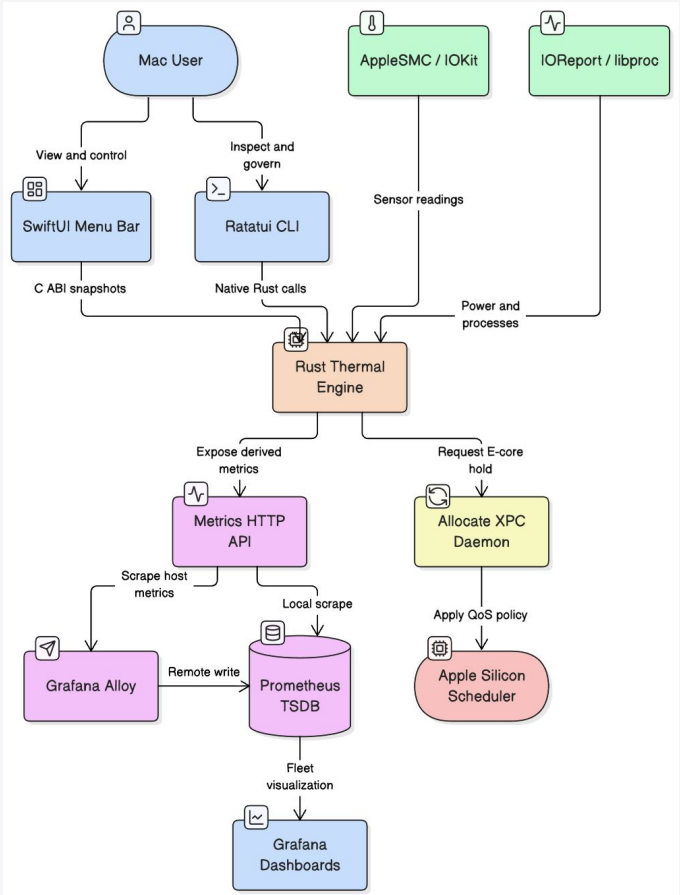
Albedo | Fleet Thermal Control Agent [GitHub]

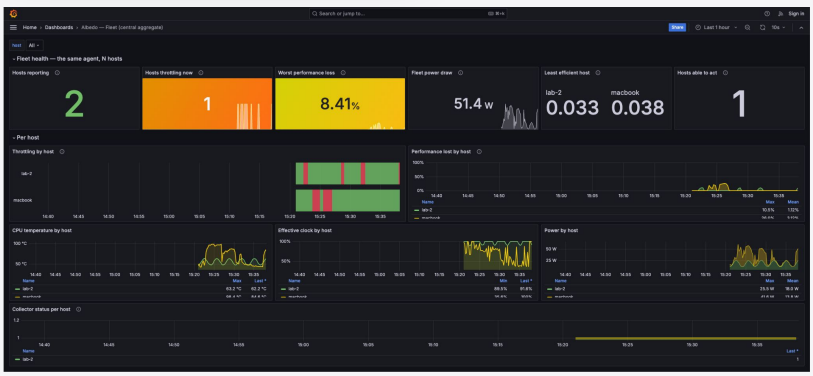
AppleSMC, IOKit | Achieved 12°C/22°F Fanless CPU Cooling | 18 Local Hardware Sensors





Grafana Individual edge device view



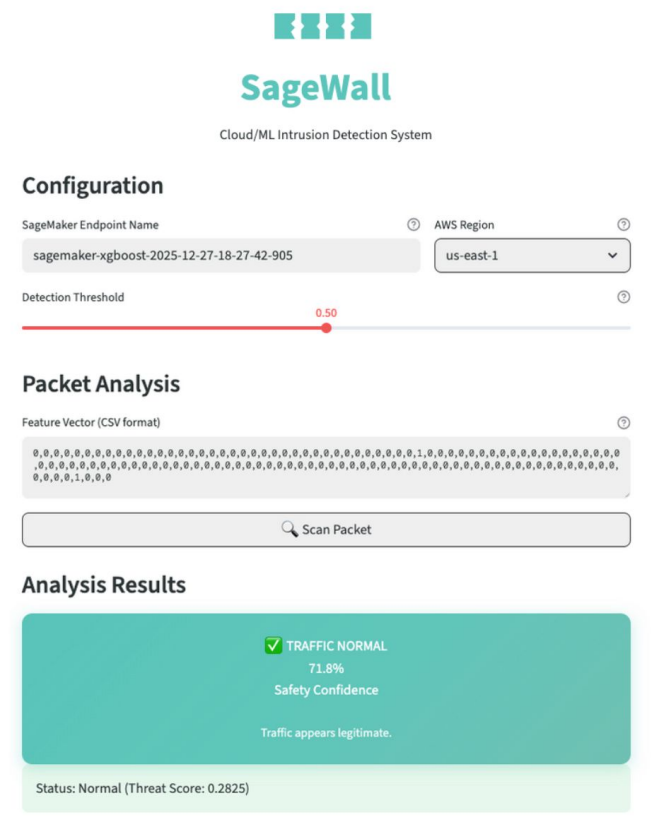


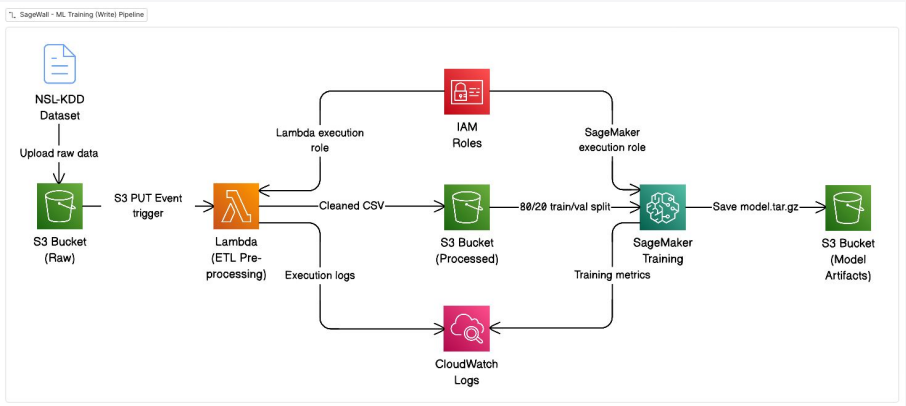
Grafana fleet control host view

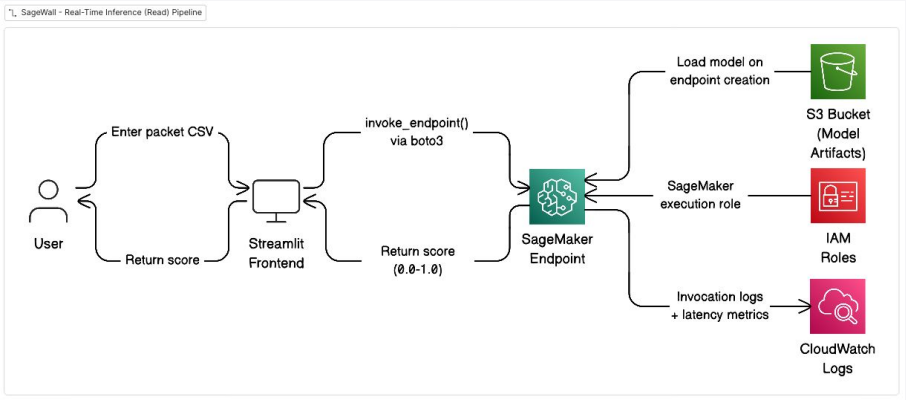
SageWall | Cloud Infrastructure IDS [GitHub] [Devlog]

AWS, Python | Automated ETL Pipeline | 99.9%* Accuracy (XGBoost) | <100ms latency

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[GitHub] [Website]







CrawlStars | Distributed Search Engine [GitHub] [Devlog]

Golang, MongoDB | O(1) lookup time | 2000+ URLs/minute | <15MB memory usage

